

STAT 514 Practice Final Solutions

Source: `midterms/midterm3/practice.pdf`. These notes emphasize reusable formulas and exam-ready calculations.

Problem 1

- (a) A 2_{IV}^{8-3} design has

$$2^{8-3} = 32$$

runs and 8 two-level factors.

- (b) In a resolution IV design, a two-factor interaction is not aliased with a main effect, but it may be aliased with another two-factor interaction.

- (c) A 2_{III}^{9-5} design has $2^{9-5} = 16$ runs, so it can estimate

$$16 - 1 = 15$$

factorial effects besides the intercept. Since $p = 5$, each effect is aliased with

$$2^p - 1 = 2^5 - 1 = 31$$

other factorial effects.

- (d) The fundamental principles for factorial effects are sparsity, hierarchy, and heredity.

- (e) A CCD using a 2_V^{5-1} factorial portion and 3 center points has

$$n_F + 2k + n_C = 2^{5-1} + 2(5) + 3 = 16 + 10 + 3 = 29$$

runs.

- (f) A 3^2 full factorial design has $3^2 = 9$ runs and 2 factors.

- (g) A Latin hypercube design with n runs in k factors has 1 point in each of the n equally sized intervals of every one-dimensional marginal, so each level is sampled 1 time per factor.

- (h) A maximin LHD maximizes the minimum pairwise distance. A minimax design minimizes the maximum distance from any candidate input to its nearest design point.

Problem 2

General rule:

$$df_E = N - p,$$

where p includes the intercept and all fitted model parameters.

(a) CRD with three treatments and 7, 8, 9 replicates:

$$N = 24, \quad df_E = N - a = 24 - 3 = 21.$$

(b) Paired comparison with 15 pairs:

$$df_E = n_{\text{pairs}} - 1 = 15 - 1 = 14.$$

(c) Full 2^4 factorial with 3 replicates and all effects:

$$N = 2^4(3) = 48, \quad p = 2^4 = 16, \quad df_E = 48 - 16 = 32.$$

Equivalently, $df_E = 2^k(n - 1) = 2^4(2) = 32$.

(d) Unreplicated 2^{8-2} design:

$$N = 2^6 = 64.$$

The model has

$$p = 1 + 8 + \binom{8}{2} = 1 + 8 + 28 = 37.$$

Thus

$$df_E = 64 - 37 = 27.$$

(e) CCD using 2^{3-1} factorial runs and 5 center runs:

$$N = 2^{3-1} + 2(3) + 5 = 4 + 6 + 5 = 15.$$

A standard second-order model for $k = 3$ has

$$p = 1 + k + k + \binom{k}{2} = 1 + 3 + 3 + 3 = 10.$$

Thus

$$df_E = 15 - 10 = 5.$$

(f) Full 3^2 factorial with two replicates:

$$N = 3^2(2) = 18, \quad p = 9,$$

so

$$df_E = 18 - 9 = 9.$$

Problem 3

- (a) Use a 16-run 2^{7-3} fractional factorial design, preferably maximum resolution IV. This is appropriate for screening seven two-level factors under a 16-run limit. There is no stated blocking factor. Sample size:

$$N = 16.$$

- (b) Use a central composite design for the three quantitative factors because a second-order model is needed for curvature, interactions, prediction, and optimization. No stated blocking factor. General sample size:

$$N = 2^3 + 2(3) + n_C = 14 + n_C.$$

With five center runs:

$$N = 19.$$

- (c) Use a space-filling design, preferably a maximin Latin hypercube design, for 12 simulator inputs. The simulator is deterministic, so replication is wasteful. No blocking factor. Since the scientists want 100 uniformly spaced levels per parameter:

$$N = 100.$$

- (d) Use a completely randomized design. The aquaria are essentially homogeneous and there is no obvious nuisance factor. No blocking factor. Balanced allocation gives:

$$N = 40, \quad 10 \text{ aquaria per diet.}$$

- (e) Use a randomized complete block design with production day as the block. Each of the three treatment levels is run within each of five days:

$$N = 3(5) = 15.$$

Problem 4

(a)

Let $i = 1, 2, 3$ index cutting speed, $j = 1, 2, 3$ index coolant type, and $k = 1, \dots, 4$ index replicates. The two-factor ANOVA model with interaction is

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \stackrel{iid}{\sim} N(0, \sigma^2).$$

The constraints are

$$\begin{aligned} \sum_{i=1}^3 \alpha_i &= 0, & \sum_{j=1}^3 \beta_j &= 0, \\ \sum_{i=1}^3 (\alpha\beta)_{ij} &= 0 \quad \text{for each } j, & \sum_{j=1}^3 (\alpha\beta)_{ij} &= 0 \quad \text{for each } i. \end{aligned}$$

(b)

There are $N = 3 \cdot 3 \cdot 4 = 36$ observations. Degrees of freedom are

$$df_A = 2, \quad df_B = 2, \quad df_{AB} = 4, \quad df_E = 3 \cdot 3(4 - 1) = 27, \quad df_T = 35.$$

Given $R^2 = 0.94$ and $SSE = 18$,

$$0.94 = 1 - \frac{18}{SST} \Rightarrow SST = \frac{18}{0.06} = 300.$$

Thus

$$MSE = \frac{18}{27} = \frac{2}{3}.$$

Cutting speed has $F = 90$:

$$MS_A = 90 \left(\frac{2}{3} \right) = 60, \quad SS_A = 2(60) = 120.$$

Coolant type has $SS_B = 90$:

$$MS_B = 45, \quad F_B = \frac{45}{2/3} = 67.5.$$

Interaction:

$$SS_{AB} = 300 - 120 - 90 - 18 = 72, \quad MS_{AB} = \frac{72}{4} = 18, \quad F_{AB} = \frac{18}{2/3} = 27.$$

Source	df	SS	MS	F
Cutting speed	2	120	60	90
Coolant type	2	90	45	67.5
Interaction	4	72	18	27
Error	27	18	$2/3$	
Total	35	300		

Problem 5

For a one-replicate 2^k factorial, the effect of term T is

$$\widehat{\text{effect}}_T = \bar{y}_{T+} - \bar{y}_{T-} = \frac{1}{2^{k-1}} \sum_{\text{runs}} s_T y,$$

where s_T is the product of coded signs for the term.

(a) Main effect of A :

$$\bar{y}_{A+} = \frac{72 + 85 + 79 + 80}{4} = 79, \quad \bar{y}_{A-} = \frac{98 + 87 + 99 + 87}{4} = 92.75.$$

$$\hat{A} = 79 - 92.75 = -13.75.$$

(b) For AB , $s_{AB} = x_A x_B$:

$$\bar{y}_{AB+} = \frac{98 + 85 + 99 + 80}{4} = 90.5, \quad \bar{y}_{AB-} = \frac{72 + 87 + 79 + 87}{4} = 81.25.$$

$$\widehat{AB} = 90.5 - 81.25 = 9.25.$$

(c) Main effect plot for A : plot $(-1, 92.75)$ and $(+1, 79.00)$. The line decreases, so high A lowers roughness.

For the AB interaction plot:

B	$\bar{y}(A = -1)$	$\bar{y}(A = +1)$
-1	$(98 + 99)/2 = 98.5$	$(72 + 79)/2 = 75.5$
+1	$(87 + 87)/2 = 87.0$	$(85 + 80)/2 = 82.5$

The lines are not parallel; the effect of A is stronger when $B = -1$.

(d) Reduced model:

$$y = \beta_0 + \beta_A x_A + \beta_{AB} x_A x_B + \varepsilon.$$

For an orthogonal two-level factorial:

$$\hat{\beta}_0 = \bar{y}, \quad \hat{\beta}_T = \frac{\widehat{\text{effect}_T}}{2}.$$

Thus

$$\hat{\beta}_0 = \frac{98 + 72 + 87 + 85 + 99 + 79 + 87 + 80}{8} = 85.875,$$

$$\hat{\beta}_A = -13.75/2 = -6.875, \quad \hat{\beta}_{AB} = 9.25/2 = 4.625.$$

Fitted model:

$$\hat{y} = 85.875 - 6.875x_A + 4.625x_A x_B.$$

(e) Predictions:

A	B	$x_A x_B$	$\hat{y}$
-1	-1	+1	97.375
+1	-1	-1	74.375
-1	+1	-1	88.125
+1	+1	+1	83.625

The minimum predicted roughness is at $A = +1, B = -1$, with $\hat{y} = 74.375$.

Problem 6

(a) The column D equals ABC , so

$$D = ABC, \quad I = ABCD.$$

(b) The column E equals $-AC$, so

$$E = -AC, \quad I = -ACE.$$

(c) The independent words are $I = ABCD$ and $I = -ACE$. Multiplying them gives

$$(ABCD)(-ACE) = -BDE.$$

Therefore the complete defining relation is

$$I = ABCD = -ACE = -BDE.$$

(d) The shortest nonidentity words are ACE and BDE , both of length 3. The design has resolution III.

(e) Reverse the signs of column B . If the reversed column is still called B , then

$$D = -ABC, \quad E = -AC.$$

Thus

$$I = -ABCD, \quad I = -ACE.$$

Multiplying gives BDE , so the complete defining relation is

$$I = -ABCD = -ACE = BDE.$$

Problem 7

(a)

$$\bar{y}_{A+} = \frac{32 + 45}{2} = 38.5, \quad \bar{y}_{A-} = \frac{54 + 47}{2} = 50.5.$$

$$\hat{A} = 38.5 - 50.5 = -12.$$

(b)

$$\bar{y}_{AB+} = \frac{54 + 45}{2} = 49.5, \quad \bar{y}_{AB-} = \frac{47 + 32}{2} = 39.5.$$

$$\widehat{AB} = 49.5 - 39.5 = 10.$$

(c) Estimate pure error from the center points 40, 38, 42:

$$\bar{y}_C = 40,$$

$$\hat{\sigma}^2 = MSE_{\text{pure error}} = \frac{(40 - 40)^2 + (38 - 40)^2 + (42 - 40)^2}{3 - 1} = 4.$$

Degrees of freedom:

$$df_E = 3 - 1 = 2.$$

(d) For a one-replicate 2^2 factorial,

$$SE(\widehat{\text{effect}}) = \sqrt{MSE}.$$

Thus

$$SE(\widehat{AB}) = \sqrt{4} = 2, \quad t_0 = \frac{10}{2} = 5.$$

With $df = 2$, $t_{0.025,2} = 4.303$. Since $|5| > 4.303$, the AB interaction is significant at the 5% level.

(e) Curvature compares center and factorial means:

$$\bar{y}_F = \frac{54 + 47 + 32 + 45}{4} = 44.5, \quad \bar{y}_C = 40.$$

$$\hat{C} = \bar{y}_C - \bar{y}_F = -4.5.$$

Standard error:

$$SE(\hat{C}) = \sqrt{MSE \left(\frac{1}{n_C} + \frac{1}{n_F} \right)} = \sqrt{4 \left(\frac{1}{3} + \frac{1}{4} \right)} = \sqrt{\frac{7}{3}} = 1.528.$$

Test statistic:

$$t_0 = \frac{-4.5}{1.528} = -2.95.$$

Since $|-2.95| < t_{0.025,2} = 4.303$, curvature is not significant at the 5% level.

Equivalent:

$$SS_{\text{curv}} = \frac{n_F n_C}{n_F + n_C} (\bar{y}_F - \bar{y}_C)^2 = \frac{4 \cdot 3}{7} (4.5)^2 = 34.714,$$

$$F_{\text{curv}} = \frac{34.714}{4} = 8.679 = t_0^2.$$

Problem 8

The fitted model is

$$y = 8 + 4x_1 - 2x_2 - 2x_1^2 - x_2^2 + \varepsilon.$$

(a) A circumscribed CCD with $k = 2$, $\alpha = \sqrt{2}$, and three center points has:

factorial points: $(-1, -1), (1, -1), (-1, 1), (1, 1)$,

axial points: $(\sqrt{2}, 0), (-\sqrt{2}, 0), (0, \sqrt{2}), (0, -\sqrt{2})$,

center points: $(0, 0), (0, 0), (0, 0)$.

Total:

$$N = 4 + 4 + 3 = 11.$$

(b) In

$$y = \beta_0 + x'b + x'Bx + \varepsilon, \quad x = (x_1, x_2)',$$

we have

$$\beta_0 = 8, \quad b = \begin{pmatrix} 4 \\ -2 \end{pmatrix},$$

and, because there is no interaction term,

$$B = \begin{pmatrix} -2 & 0 \\ 0 & -1 \end{pmatrix}.$$

(c) The stationary point solves

$$b + 2Bx_s = 0, \quad x_s = -\frac{1}{2}B^{-1}b.$$

Since

$$B^{-1} = \begin{pmatrix} -1/2 & 0 \\ 0 & -1 \end{pmatrix},$$
$$B^{-1}b = \begin{pmatrix} -2 \\ 2 \end{pmatrix}, \quad x_s = -\frac{1}{2} \begin{pmatrix} -2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The stationary point is $(x_1, x_2) = (1, -1)$.

(d) The eigenvalues of the diagonal matrix B are

$$\lambda_1 = -2, \quad \lambda_2 = -1.$$

Both are negative, so B is negative definite and the stationary point is a maximum. The predicted maximum response is

$$\hat{y}(1, -1) = 8 + 4(1) - 2(-1) - 2(1)^2 - (-1)^2 = 11.$$